

Wh*-infinitives as topic-setting questions

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1. Goals and main claims

This paper investigates the semantics and pragmatics of root *wh*-infinitive questions (INFQs); i.e., unembedded *wh*-questions featuring an infinitival verb, as in examples (1) and (2).

- (1) [Two friends are looking over the menu at a restaurant. One says to the other:]
(Okay,) What to order?
- (2) [Discussing journals with your co-author]
(Hmm,) Where to send our manuscript?

Building on prior work on non-finite *wh*-constructions (Kjellmer 1982, Duffley and Enns 1996, Bhatt 1999, Gärtner 2017), we propose that the INFQs in (1) and (2) should be characterized as *topic-setting questions*, a term we borrow from Farkas (2025). Informally, a topic-setting question is a non-canonical speech act in which the Speaker assumes that no salient answer is currently available to resolve the issue raised by the question. The conversational function of uttering a topic-setting question is to prompt deliberation in order to arrive at a resolution. We argue that this is a conventionalized effect that arises from the marked status of INFQs relative to canonical finite *wh*-questions.

This paper proceeds as follows. §2 outlines the core properties of INFQs. §3 provides a compositional analysis of the semantics of INFQs, and shows that a purely semantic account does not capture the pragmatic profile in §2. §4 argues that the topic-setting function of INFQs follows from their status as marked, insubordinate constructions (Evans 2007). §5 highlights avenues for future research, and concludes.

For reasons of space, we set aside non-information seeking *wh*-infinitives (dubbed *af-firmative* questions in Duffley and Enns (1996)), which often function as titles, such as the example in (3). See Duffley and Enns (1996) for more on this kind of construction.

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- (3) What to Do When the Taxman Comes: the Inside Story on How to Cope with Canada's Tax Department (Ferguson 1979, book title. Cited in Duffley and Enns 1996:228)

2. Core properties of INFQs

This section lays out several syntactic, semantic, and pragmatic properties of INFQs that form the empirical basis for the analysis developed in Sections 3 and 4.

2.1 Syntactic and semantic properties

In terms of their syntax and semantics, non-finite *wh*-clauses pattern like *questions*.¹ For instance, as discussed extensively in Chapter 4 of Bhatt (1999), they can embed under so-called *rogative* and *responsive* predicates that select for questions, as in (4), and they cannot be embedded under *antirogative* predicates, those predicates that exclusively select declarative complements, as in (5).² Furthermore, as shown in example (6), INFQs can be coordinated with canonical finite questions.

- (4) {I asked/I know/the question is} where to go next.

- (5) *{I believe/I am certain/I doubt} where to go next.

- (6) Where to go next and how are we getting there?

However, as previously noted in Duffley and Enns (1996), the range of *wh*-expressions that appear in INFQs is a subset of those that occur in finite *wh*-questions and embedded *wh*-infinitives; in particular, while *who*, *what*, and D-linked *wh*-phrases featuring *which* (7a), as well as *where*, *when*, and *how* (7b) freely occur in INFQs, *why* cannot (7c).^{3,4}

- (7) a. (Hmm,) {**what / who / which dish**} to bring to the party?
 b. (Hmm,) {**where / when / how**} to throw the party?
 c. #(Hmm,) **why** to throw the party?

We also note that *wh*-word in an INFQ cannot be the grammatical subject of the clause.

¹We assume that, syntactically, *wh*-interrogative clauses are CPs headed by an interrogative C element, with *wh*-phrases moving to [Spec,CP]. Our semantic assumptions are spelled out in §3.

²See, e.g., Grimshaw (1979), Lahiri (2002), Theiler et al. (2017), Uegaki and Sudo (2017) for more on the distinction between these three types of clause-embedding predicates.

³See Bhatt (1998) on an alternate construction that consists of *why* and a bare non-finite verb (as in *Why throw the party?*). Bhatt classifies these bare *why* questions as *rhetorical questions*.

⁴Kjellmer (1982:452) claims that polar infinitival questions featuring *whether*, such as (i), are impossible, however, we find them to be acceptable, if marked, constructions.

- (i) ?(Hmm,), **Whether** to throw the party?

- (8) #Who to teach semantics next term? $\neq$ Who should teach semantics next term?

INFQs are obligatorily modal (Bhatt 1999). They receive a circumstantial interpretation, often with a bouletic or teleological flavour (following Portner's (2009) framework), as in (9a). They do not receive epistemic readings, as shown in (9b).

- (9) What to teach next year?
a. $\approx$ What {could/should} I teach next year?
b. $\neq$ What {will/might} I teach next year?

INFQs are also obligatorily future-oriented. This is evidenced by the contrast in (10a–10b): while an INFQ is incompatible with the past-oriented adverbial *yesterday* in (10a), a finite *wh*-question containing a modal freely permits a past-oriented interpretation, as shown in (10b).

- (10) a. #What to teach yesterday?
b. What should I have taught yesterday?

2.2 Discourse effects of INFQs

Following Farkas (2022), canonical questions are associated with the assumptions in (11), which are inspired by traditional speech act theory (e.g., Searle 1969).

- (11) Default assumptions accompanying question acts (Farkas 2022)
- a. **Speaker ignorance:** The speaker's epistemic state is neutral relative to the possible resolutions of the issue she raises.
 - b. **Addressee competence:** The speaker assumes that the addressee knows the information that settles the issue she raises.
 - c. **Addressee compliance:** The speaker assumes that the addressee will provide this information in the immediate future of the conversation as a result of the speaker's speech act.
 - d. **Issue resolution goal:** It is assumed that the main aim the speaker pursues when raising an issue is to have it resolved in the immediate future of the conversation.

Non-canonical questions (e.g., biased and rhetorical questions) arise when one or more of the default assumptions in (11) are weakened or suspended. We argue that INFQs function as *topic-setting questions*: a type of non-canonical question in which the assumption of Addressee competence is suspended and in which the responder is taken to be the group consisting of the discourse participants (Farkas 2025:141,162).⁵

INFQs pattern with canonical questions in most respects. In particular, they assume Speaker ignorance, Addressee compliance, and the Issue resolution goal. However, INFQs

⁵Farkas (2025:141) characterizes the (non-INFQ) question in (i) as a *topic-setting question*:

differ from canonical questions in that the assumption of Addressee competence is either weakened or suspended (henceforth *suspended*).⁶ For example, consider the following exchange between Allie and Ben who are organizing a last-minute dinner party and have already settled most of the details except for what to serve.

- (12) Allie: We still haven't figured out what we're going to cook tonight.
Ben: Right. Okay, what to cook for dinner?

In (12), neither Allie nor Ben is assumed to possess the information that would immediately settle the issue of *what to cook*. By uttering the INFQ in (12), Ben signals that they should work together toward jointly resolving the issue; we take this to mean that, in spite of the suspension of Addressee competence, both Addressee compliance and the Issue resolution goal are taken to hold in the INFQ in (12).

Further evidence for the suspension of Addressee competence comes from the following pair of examples. As the example in (13a) shows, it is felicitous to follow up a canonical (finite) question with a statement asserting that the addressee knows the answer to the issue raised by the question. In contrast, such a continuation is infelicitous with an INFQ, as illustrated in (13b).

- (13) a. Who will we invite to give the keynote? I know you know!
b. Who to invite to give the keynote? #I know you know!

These special discourse effects are a root-level phenomenon. Embedded infinitival *wh*-questions do not display the same suspension of Addressee competence, as shown in the pair in (14).

- (14) [A detective interrogating a jewel thief says to the thief:]
a. Tell me where to find the stolen jewels!
b. #Where to find the stolen jewels?

At the same time, INFQs are distinct from another class of non-canonical questions that suspends addressee competence, namely *conjectural questions* (Littell et al. 2010, Eckardt 2020). Unlike conjectural questions, approximated in English by questions embedded under *I wonder* in (15a), INFQs/topic-setting questions are not used simply to express curiosity or speculate aloud without directing the issue toward the addressee.

- (15) [Realizing after a while that you got on the wrong bus]

- (i) [A linguist to her colleague, when setting down to work together]
What does our analysis predict? That's what we have to figure out first.

⁶Farkas (2022:297) proposes that Addressee compliance entails Addressee competence, which is at odds with our classification of INFQs, where Addressee competence is suspended while Addressee compliance is maintained. On our view, Addressee compliance does not require the speaker to assume that the addressee already has the information needed to resolve the issue. Rather, it is associated with the expectation that the addressee will act cooperatively in jointly working toward the resolution of the issue.

- a. I wonder where we might be going. . . (Conjectural question)
- b. #Where to go? (INFQ)

We suggest that a crucial difference between conjectural questions and INFQs concerns the Issue resolution goal. Following Farkas (2022), after Eckardt (2020), conjectural questions suspend this assumption: they raise an issue without assuming it will be resolved. By contrast, INFQs retain the Issue resolution goal despite suspending Addressee competence. Consequently, unlike conjectural questions, INFQs expect a response.⁷

3. The semantics of INFQs

This section presents a compositional account of INFQs. We account for the semantic effects of these questions, but this alone does not derive their pragmatic profile. We assume a standard Hamblin-Karttunen semantics for interrogatives, under which questions denote sets of propositions corresponding to their possible answers.

INFQs are *partial-control* infinitives in the sense of Landau (2000). In these root-clause contexts, we take the PRO subject to be interpreted by default either as the speaker alone or as a group containing the speaker and additional contextually determined participants. This is illustrated by the examples in (16). Collective predicates such as (intransitive) *meet* require a plural subject, which rules out (16a). However, the embedded infinitival question in (16b) is acceptable because the PRO subject receives a partial-control interpretation as a group containing both the speaker and the board. Finally, (16c) shows that the same plural interpretation is available in INFQs when supported by the discourse context.

- (16)
- a. #**I** will meet.
 - b. The board informed **me** [when PRO to meet next].
 - c. [We are adjourning our meeting, I ask the chair:]
When PRO to meet next?

To represent this, we assume that for any index i , $\llbracket PRO_{Sp+i} \rrbracket^g$ denotes the mereological sum of the speaker in the context Sp and $g(i)$:⁸

$$(17) \quad \llbracket PRO_{Sp+i} \rrbracket^g = Sp \oplus g(i)$$

In an INFQ such as *what to eat?*, *eat* first applies to the variable introduced by the *wh* trace $g(1)$ and PRO_{Sp+i} :

⁷ Another important difference that is worth pursuing concerns the flavour of modality typically associated with conjectural questions (epistemic) vs. INFQs (circumstantial).

⁸ This denotation is only applicable to PRO in root INFQs, not embedded infinitival questions, as the latter has PRO co-indexed with the subject of the matrix clause, not necessarily the speaker. We also set aside the question of why INFQs involve partial-control, though we speculate that this feature is inherited from their embedded analogues—as noted in Landau (2000:15), question embedding predicates form a subset of the predicates that invoke partial-control.

$$(18) \quad \llbracket PRO_{Sp+i} \text{ to eat } t_I \rrbracket^g = \lambda t \lambda w. \mathbf{eat}(g(1))(Sp \oplus g(i))(t)(w)$$

To capture the obligatory future orientation of INFQs discussed in §2.1, we adopt a version of Abusch’s (1997) WOLL, following Wurmbrand (2004) in treating future orientation as introduced within the infinitival clause itself. WOLL existentially closes the event time associated with the vP and introduces a new reference time t' that precedes it.

$$(19) \quad \llbracket \text{WOLL } PRO_{Sp+i} \text{ to eat } t_I \rrbracket^g = \lambda t' \lambda w. \exists t [t' < t \wedge \mathbf{eat}(g(1))(Sp \oplus g(i))(t)(w)]$$

We further propose that the modal component of INFQs is introduced in the specifier of the infinitival interrogative complementizer (here, $C_{[+Q]}$), following Bhatt (1999). We implement a Kratzer-style modal semantics using BEST, which evaluates propositions relative to a conversational background f which is comprised of a modal base and an ordering source (Kratzer 1977, 1981). Crucially, the function in (20) is only defined if the modal base is circumstantial, accounting for the restriction against epistemic readings.

Finally, the interrogative complementizer $C_{[+Q]}$ converts the proposition into a set of propositions, while the *wh*-expression binds the variable introduced by the trace.

$$(20) \quad \llbracket \text{what } C_{[+Q]} \text{ WOLL } PRO_{Sp+i} \text{ to eat } t_I \rrbracket^{f,g} \\ = \{p : \exists x [p = \lambda w. \exists w' \in \text{BEST}(f(w)) : \exists t [t_c < t \wedge \mathbf{eat}(x)(Sp \oplus g(i))(t)(w)]]\}$$

In prose, this denotation describes a set of propositions that vary with respect to some individual x eaten by PRO at a time following the utterance time in circumstantially optimal worlds. However, this denotation alone does not explain the discourse behaviour of INFQs. Their pragmatic profile therefore requires an additional account beyond their truth-conditional semantics.

4. Modelling the discourse effects of INFQs

4.1 The Table Model

We model the discourse effects of uttering an INFQ using the Table Model (Farkas and Bruce 2010), and adopt the following basic semantic assumptions: declarative and interrogative clauses denote sets of propositions, which are referred to as *issues* I . Declaratives denote singleton sets, whereas interrogatives denote non-singleton sets. The informative content of I , $\text{Info}(I)$, is the union of all propositions in I ($\cup I$) (Farkas 2022:302–303).

The Table Model consists of the following components (these simplified definitions are adapted from Farkas 2022 and Woods and Haegeman 2023. See Farkas (2022:304–305) for details). The **Discourse commitments** of each discourse participant x (DC_x) are the set of propositions that a discourse participant x has publicly committed to in the current conversation. **The Table** is where active issues are placed to be resolved. An issue is considered resolved if some proposition from that issue is accepted into the *common ground* (Stalnaker 1978; i.e., the shared commitments of the discourse participants). **The projected set (ps)** is a set of preferred discourse commitments as envisioned by the speaker upon making an

utterance. For each alternative p in an issue I , the ps contains a commitment list of the form $DC_x \cup \{p\}$, where x is contextually determined and defaults to the addressee (Farkas 2022, adapted from Meriçli 2016).

When a canonical *wh*-question is uttered, the issue is placed on the Table, the informative content of the issue, which corresponds to the set of possible answers $p_1, p_2, p_3, \dots$ is added to the discourse commitments of the speaker, and the projected set is generated as the set of possible addressee commitments to each proposition in I . This has the effect of putting the addressee “on the spot”, to commit to one of the alternative propositions in p as a first step toward resolving the issue.

- (21) Conversational state following the utterance of a *wh*-interrogative

DC _{Sp}	Table	DC _{Ad}
Info(I)	$\{p_1, p_2, p_3, \dots\}$	
	ps: $\{DC_{Ad} \cup \{p_1\}, DC_{Ad} \cup \{p_2\}, DC_{Ad} \cup \{p_3\}, \dots\}$	

4.2 INFQs as marked questions

Turning back to INFQs, we propose that their discourse effects established in §2.2 stem from their status as marked interrogative constructions in comparison to unmarked, finite *wh*-questions. This markedness arises from their *insubordinate* structure—they are root-level utterances which feature subordinate-clause syntax (Evans 2007). We represent this markedness relation in (22).

- (22) finite *wh*-questions < INFQs

Cross-linguistically, root non-finite clauses are commonly associated with conventionalized expressive or non-canonical discourse functions. English infinitival exclamatives provide one familiar example.

- (23) Oh, to be in England
 Now that April’s there!
 (Robert Browning, *Home Thoughts from Abroad*, cited in Jespersen 1933, Kjellmer 1982)

We propose that INFQs also involve a conventionalized special discourse effect beyond their ordinary question denotation established in §3. Following Farkas’s (2025) analysis of rhetorical questions, we treat this effect as a *mandatory conversational implicature* (CI) in the sense of Lauer (2014). The implicature associated with INFQs is given in (24).

- (24) **Immediate public goal when asking an INFQ with propositional content I :**
 When asking an INFQ, Speaker’s immediate public goal is to collectively resolve the issue I .

We further argue that the special discourse effect associated with INFQs is reflected in the structure of the projected set. For canonical *wh*-questions, the projected set consists of a list of conversational moves in which the addressee commits to one of the propositions in the issue, thereby taking the first step toward resolving it. By contrast, the CI in (24) specifies that the immediate public goal of an INFQ is the collective resolution of the issue it raises. We can model this difference in the Table model by altering the structure of the projected set.⁹ INFQs project discourse states in which both discourse participants commit to the same proposition. In this way, the marked form of an INFQ conventionally directs the conversation toward futures in which resolution of the issue is a joint undertaking of speaker and addressee. This difference is represented in Table 25. As with ordinary questions, the issue itself is placed on the Table and added to the speaker's discourse commitments; however, the projected set now consists of discourse states in which both participants commit to a proposition in *I*.

(25) Conversational state following the utterance of an INFQ

DC_{Sp}	Table	DC_{Ad}
Info(<i>I</i>)	$\{p_1, p_2, p_3, \dots\}$	
	ps: $\{DC_{Sp} \cup DC_{Ad} \cup \{p_1\}, DC_{Sp} \cup DC_{Ad} \cup \{p_2\}, \dots\}$	

This analysis captures the observation that INFQs are used in collaborative or deliberative contexts, where both discourse participants are expected to contribute toward resolving the issue. It also explains why INFQs are infelicitous in contexts where the speaker assumes that the addressee can independently provide the answer, as established in §2.2.

5. Taking stock

We have argued that INFQs function as topic-setting questions, and characterized them, following following Farkas's (2025:162) definition of topic-setting questions, as belonging to the broader class of non-canonical questions that weaken or suspend the assumption of Addressee competence that is otherwise associated with canonical questions (Farkas 2022, Woods and Haegeman 2023). Unlike ordinary information-seeking questions, INFQs do not assume that the addressee already knows the answer. Instead, they signal that the discourse participants should jointly deliberate toward resolving the issue under discussion.

The analysis proposed here derives this effect from the interaction of the semantics and pragmatics of the construction. Semantically, INFQs denote future-oriented circumstantial questions involving unrealized possibilities or potential courses of action. Their modal interpretation directs attention toward decisions, plans, and possible future outcomes. At the same time, the partial-control interpretation of PRO strongly favours readings involving the speaker, either alone or as part of a larger group. On the pragmatic side, we proposed that the marked status of INFQs as in subordinate structures gives rise to a conventionalized

⁹This approach requires that CIs are in some sense present in the Table model, which is consistent with their characterization in Farkas (2025).

conversational implicature: namely, that the issue should be resolved collectively by the discourse participants.

An issue we have left unaddressed concerns self-addressed INFQs, such as in (26).

- (26) [Allie's neighbour gave her a kohlrabi from his garden. She's never tried one before. Later, alone in her kitchen, she says to herself:]
(Okay,) How to eat a kohlrabi?

Exactly how self-addressed INFQs should be integrated into the present proposal remains a topic for future research. However, we hypothesize that INFQs naturally lend themselves to such self-addressed contexts due to the suspension of the assumption that another discourse participant is in a position to resolve the issue.

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